

Midterm 1

Time Limit: 50 minutes

- **Every non-trivial calculation and claim in the exam must be properly justified.**
- Your work must be well organized and coherent. Pay attention to notation, definitions of events, units of measurement, and interpretation.
- **You must answer each question inside the designated box;** answers outside will not be considered.
- The use of calculators is allowed.
- Digital devices such as mobile phones, tablets, and laptops, as well as internet access, are forbidden.

The following questions are about a fictional city, *Valverde*, which is considering a nonpartisan civic information campaign before a local election. The city first studies survey patterns, then evaluates statistical evidence about support for the campaign, and finally considers whether the campaign has a causal effect on civic knowledge.

1. (2.5 points) The city surveys 500 residents. For each resident, it records whether the resident is under 30 years old and whether the resident supports sending nonpartisan election reminders by text message.

	Supports reminders (S)	Does not support reminders (S^c)
Under 30 (Y)	132	68
30 or older (Y^c)	168	132

- (a) Compute $P(Y)$, $P(S)$, $P(Y \cap S)$, and $P(S | Y)$.

- (b) Are Y and S independent? Justify your answer using probabilities.

- (c) Let R be the event that a resident follows local politics closely. Suppose $P(R | S) = 0.70$ and $P(R | S^c) = 0.40$. Use the Law of Total Probability to compute $P(R)$.

- (d) If a resident follows local politics closely, what is the probability that the resident supports reminders? Interpret this as an update from $P(S)$.

2. (2.5 points) The mayor claims that a majority of residents support the election reminders. A new random sample of $n = 625$ residents finds that 350 support the reminders.

- (a) Let $\hat{p}$ be the sample proportion who support reminders. Compute $\hat{p}$. What does the Law of Large Numbers say about $\hat{p}$ as the sample size becomes large?

- (b) Write the null and alternative hypotheses for testing whether support is above 50%.

- (c) Under the null hypothesis, use

$$SE_0 = \sqrt{\frac{0.50(1 - 0.50)}{625}} = 0.020$$

to compute the test statistic. The one-sided p-value for this test statistic is about 0.0013. What do you conclude at the 5% level?

- (d) Use

$$SE(\hat{p}) = \sqrt{\frac{0.56(1 - 0.56)}{625}} \approx 0.0199$$

to compute an approximate 95% confidence interval. Does the interval agree with your test conclusion?



3. (2.5 points) Valverde also runs a randomized experiment. Residents are randomly assigned to either read a short civic information message or read a neutral message. Afterward, each resident answers a short civic-knowledge quiz scored from 0 to 100.

Group	n	Mean score	Standard deviation
Civic information message ($T = 1$)	100	68	18
Neutral message ($T = 0$)	100	62	20

- (a) Compute the estimated difference in means, $\bar{Y}_1 - \bar{Y}_0$.

- (b) Use

$$SE = \sqrt{\frac{18^2}{100} + \frac{20^2}{100}} \approx 2.69$$

to test $H_0 : \mu_1 - \mu_0 = 0$ against $H_A : \mu_1 - \mu_0 \neq 0$. The two-sided p-value is about 0.026.

- (c) Compute an approximate 95% confidence interval for the difference in means and interpret it.

- (d) Why does random assignment matter for interpreting this difference as a causal effect?



4. (2.5 points) The city is also interested in a voluntary online civic workshop. Residents choose whether to take the workshop. The outcome Y_i equals 1 if resident i says they are very likely to vote, and 0 otherwise.

Group	Number of residents	Mean of Y_i
Took the workshop ($T = 1$)	120	0.72
Did not take the workshop ($T = 0$)	180	0.58

- (a) Define $Y_i(1)$, $Y_i(0)$, and the individual treatment effect TE_i in this study.

- (b) State the fundamental problem of causal inference in this example.

- (c) Compute the difference in means between workshop participants and non-participants. Should we automatically interpret this as the causal effect of the workshop? Explain.

- (d) Propose one design change that would make the causal interpretation more credible. Mention one assumption that would still matter.